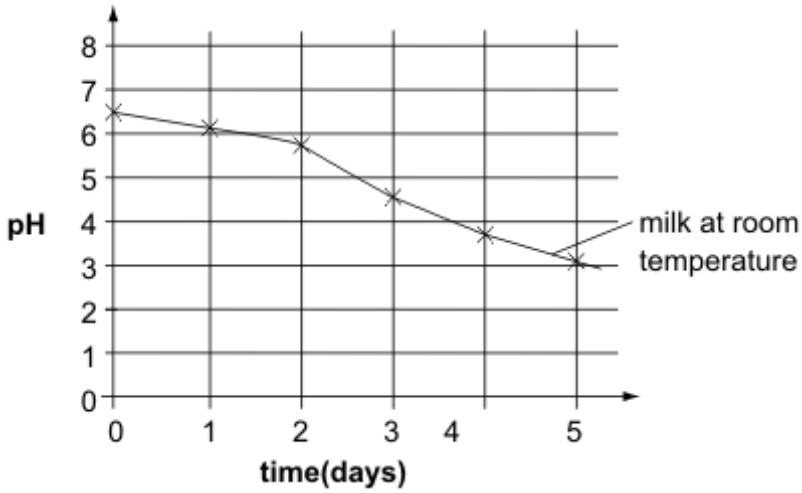
**Q1.** Jane stored some milk at room temperature for five days in a sealed container.
She used a pH sensor and data logger to record the pH of the milk for 5 days.

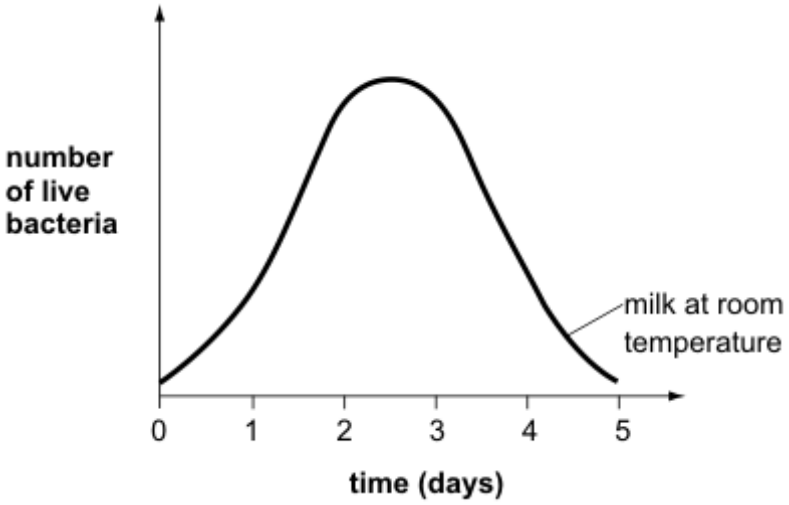
Her results are shown below.

**graph 1**




(a) Jane predicted that the number of live bacteria in the milk would change as shown below.

**graph 2**




(i) Suggest one reason why the number of live bacteria would start to decrease after 3 days.

.....

.....

1 mark

(ii) What evidence from **graph 1** suggests that there were still some live bacteria in the milk on day 5?

.....

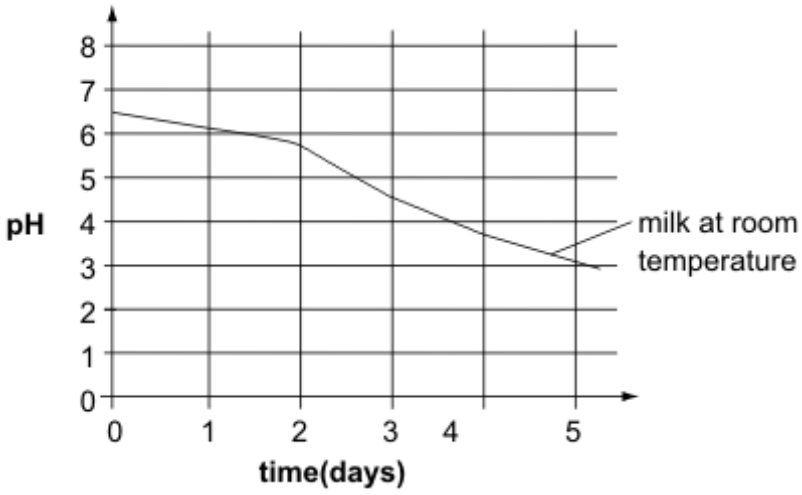
.....

1 mark

(b) Jane put some fresh milk in a sealed container in the fridge.
She measured the pH of the milk every day for five days.

(i) On **graph 3** below, draw a line to show the pH of the **refrigerated milk** for five days.

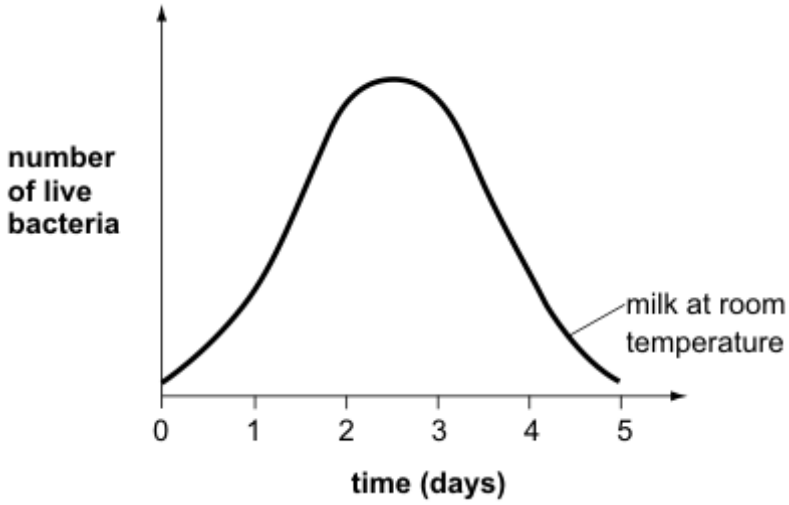
**graph 3**




2 marks

(ii) On **graph 4** below, draw a line to predict how the number of live bacteria in **refrigerated milk** will change over five days.

**graph 4**




1 mark
maximum 5 marks

## Q1.

(a) (i) any **one** from

- the pH is too low

  *'the milk goes off' is insufficient*

- it is too acidic

  *accept 'acid is killing the bacteria'*
  *'the bacteria were dying' is insufficient*
  *accept 'the bacteria could be poisoned'*
  *accept 'not enough nutrients or sugar'*
  *'there is no sugar left' is insufficient*

**1 (L7)**

(ii) the pH was still decreasing on day 5

*accept 'it was going down'*
*do **not** accept 'it has not reached 0'*
*accept 'the acidity was increasing'*
*'it has not reached the bottom of the graph' is insufficient*
*'the line continues' is insufficient*

**1 (L7)**

(b) (i)

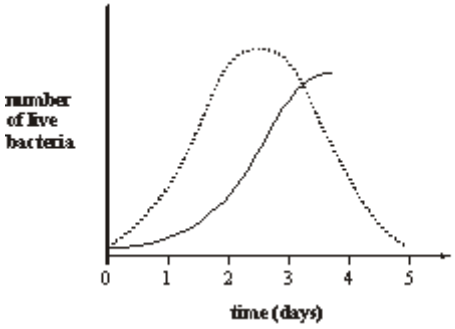
- a line that starts at pH 6.5 and then falls

**1 (L6)**

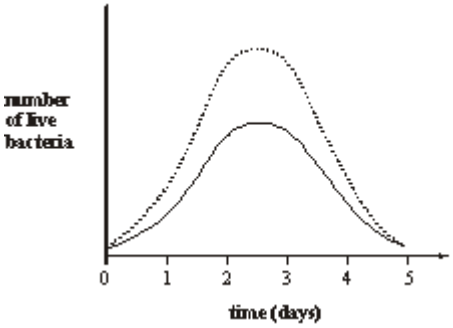
- a response that decreases less than 3.5 over 5 days

**1 (L6)**

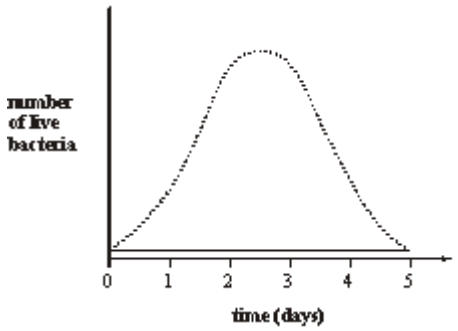
(ii) a graph that indicates that the number of bacteria increases more slowly, starting at same point




*accept a lower graph over the same line*




*the line must be below the dotted line for the first 2.5 days*
*do **not** accept a horizontal line above the x axis as shown below*




**1 (L7)**

**[5]**